MILITARY SPECIFICATION
RADIOSONDE SET AN/AMT-12( )
(Units of)

1. SCOPE

1.1 This specification covers the following units of one type of balloon borne meteorological measuring equipment, designated Radiosonde Set AN/AMT-12( ).

Transmitter, Radiosonde T-652( )/AMT-12
Modulator, Radiosonde MD-317( )/AMT-12

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-398	Plastic molding material, styrene butadien
L-P-513	Plastic sheet, laminated, thermosetting, paper base, phenolic-resin
L-T-90	Tape, Pressure-sensitive, Adhesive, (Cellophane and Cellulose Acetate)
QQ-S-365	Silver plating, electrodeposited, general requirements for

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MIL-E-1/1311	Electron Tube 7533
MIL-E-1/1460	Electron Tube 5875
MIL-R-11	Resistor, Fixed Composition; insulated, General Specification For
MIL-R-10509	Resistors, Fixed Film, High Stability, General Specification For
MIL-P-11268	Parts, Materials, and Processes used in Electronic Communications Equipment
MIL-R-11091	Radiosonde Set AN/AMT-12, Packaging and Packing of
MIL-T-12284	Temperature Element, Resistance, Met Temperature Element ML-419/AMT-4

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MIL-H-13186	Humidity Element, Resistance ML-476/AMT
MIL-M-13231	Marking for Electronic Items
MIL-W-16878	Wire, Electrical, insulated, high temperature
MIL-P-22748	Plastic, Polyethylene and Copolymers, High density, molding and extrusion material
MIL-G-45204	Gold Plating, Electro Deposited
MIL-C-45662	Calibration Systems Requirements
MIL-R-46085	Rhodium Plating, Electrodeposited
MIL-C-55076	Computer, High Temperature CP-223/UM

STANDARDS

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-D-30164	Chart, Radiosonde, Baroswitch, Calibration
SC-A-46491	Relay, Radiosonde
SC-GL-323214	Gages for Radiosonde AN/AMT-12
SC-A-46884	Hypsometer ML-606( )/AM

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. - Radiosonde Set AN/AMT-12( ) is a free balloon-borne meteorological instrument consisting of a Modulator (MD-317( )/AMT-12), a Transmitter (T-652( )/AMT-12), a Temperature Element, Resistance (ML-419/AMT), a Humidity Element, Resistance (ML-476/AMT) and a relay per SC-A-46491. The sensor elements located in and on the Modulator

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sense meteorological parameters. Through switching circuitry these parameters are sequentially introduced into the grid circuit of the blocking oscillator in the transmitter. The output of the blocking oscillator, whose rate is dependent upon meteorological parameters and a reference resistor, amplitude modulates (AM) the R.F. oscillator output of 1660 MHz - 1700 MHz for transmission of upper air meteorological data to ground station equipment, Rawin Set AN/GMD-1( ) or AN/GMD-2( ).

3.2 First Article. - The contractor shall furnish thirty (30) First Article (preproduction) samples for approval if required by the invitation for bids and contract.

3.3 Procurement Model. - A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model and the equipment shall incorporate all features of the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.4 Parts, Materials and Processes, General. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4.1 Baroswitch Contact Point. - The contact point which bears on the commutator shall be made of heat treated, hardened Beryllium-copper, gold plated with a final plating of Rhodium Type I, Class II per MIL-R-46085. (See 4.7.12)

3.4.2 Relay. - Two hermetically sealed, single-pole, double-throw, (SPDT) relays, per SC-A-46491, shall be connected into the modulator circuit in accordance with the procurement model.

3.4.3 Aneroid Capsule. - The aneroid pressure capsule shall be fabricated from "Ni Span C" alloy, aged under ambient conditions for 30 calendar days prior to calibration to insure that the capsule has attained a stabilized condition. The date of fabrication shall be legibly and indelibly marked on the capsule.

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3.4.4 Baroswitch Frame and Support. - The baroswitch frame and commutator support shall be constructed in accordance with the model to provide rigid support for the aneroid capsule, the contact arm support, the commutator and the detent mechanism.

3.4.4.1 Hypsometer Assembly. - Each part of the hypsometer assembly shall be constructed in accordance with the model and the specifications referenced in their respective sub-paragraphs of paragraph 3.9.1.3.

3.4.5 Component Mounting Boards. - The component mounting boards in the radiosonde shall be fabricated from laminated thermosetting plastic material, Type PBE-P, per Federal Specification L-P-513.

3.4.6 Case Assemblies. - The modulator and transmitter case assemblies and the temperature element support arms shall be fabricated from plastic molding compound, styrene-butadiene per Federal Specification L-P-398, Type IV, Class 2 with white opaque filler, or equal. The cases and element support arms shall not warp or become distorted under any specified service conditions, and shall be free of sharp edges and other molding irregularities. (See 4.7.2 and 4.9)

3.4.7 Wire, Hookup. - Hookup wire shall be stranded and shall conform to MIL-W-16878. (See 4.7.3)

3.4.8 Plugs and Receptacles. - Plugs and receptacles shall conform to the procurement model. (See 4.7.3)

3.4.9 Electron Tubes. -

3.4.9.1 Blocking Oscillator. - The blocking oscillator circuit shall use one each Type 5875 electron tube per MIL-E-1/1460.

3.4.9.2 R.F. Oscillator. - The R.F. oscillator circuit shall use one each Type 7533 electron tube per MIL-E-1/1311.

3.4.10 Antenna System. - The coaxial tube on which the cone reflector is mounted shall be silver plated, per Federal Specification QQ-S-365, Type 1, and then finished with clear lacquer. The cone reflector and antenna shall be brass, finished with a clear lacquer. (See 4.7.4)

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3.4.11 Resistors. - (See 4.7.14)

3.4.11.1 Fixed Composition Resistors. - Shall be in accordance with MIL-R-11 and Table I below:

TABLE I

Resistance	Resistance @27 C ±7 C	Resistance @-55 C ±5 C
2400 ohms	2400 ohms ±5%	2400 ohms ±10%
2.4 megohms	2.4 megohms ±10%	2.4 megohms ±20%

3.4.11.2 Reference Resistor. - The blocking oscillator grid (reference) resistor shall have a resistance of 112,400 ohms ±1%, with a temperature coefficient of .002% per degree centigrade over the temperature range of -55 C to +85 C, per MIL-R-10509.

3.4.12 Ancillary Items. - The ancillary items consist of one each of the following: (See 4.7.6)

a. ML-419( )/AMT, Temperature Element, Resistance; per MIL-T-12284.

b. ML-476( )/AMT, Humidity Element, Resistance; per MIL-H-13186.

c. Chart, Radiosonde Baroswitch, Calibration; per SC-D-30164.

d. Chart, Radiosonde Hypsometer, Calibration.

3.4.12.1 Ancillary Items Packaging. - The ancillary items shall be contained in a heat sealed plastic bag and shall be placed in the modulator battery compartment. (See 4.7.6)

3.4.13 Surface Treatment. - The surface of the case between the humidity element clips, and the temperature element clip and the support arms shall be so treated as to resist wetting by rain water, or manufactured of a material equal to the specified material but possessing non-wetting properties. (See 4.8.11)

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3.4.13.1 Construction for Rainproofing. - The radiosonde cases shall be constructed and rainproofed in the same manner as the procurement model such that there shall be no incursion of moisture when subjected to the test of 4.9.4.

3.4.14 Tropicalization. - (See 4.9.7)

3.4.14.1 Treatment of Materials. - Fungus-inert materials shall be used where practicable. These materials shall be in accordance with Requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with Requirement 9 of MIL-STD-454 and then shall be treated as shown in Table II below:

TABLE II

Material	Applicable Para in MIL-P-11268	Material	Applicable Para in MIL-P-11268
Cotton thread and duck	3.16	Leather	3.18
Cork	3.13	Linen thread	3.16
Felt, hair or wool	3.17	Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base	3.21

3.4.14.2 Statement of Treatment. - The statement of treatment shall indicate whether or not nutrient materials have been employed. If nutrient materials have been used, the statement of treatment shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2.1)

3.4.14.3 Fungus. - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9.8. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. (See 4.9.7)

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3.4.15 Marking. -

3.4.15.1 General. - Marking shall conform to Table II, Group II of MIL-M-13231. (See 4.4)

3.4.15.2 Visibility. - Wherever practicable, parts shall be so mounted tha their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.15.3 Serial Numbers. - The following items shall have serial numbers:

a. Modulator, Radiosonde MD-317( )/AMT-12. Baroswitch frame and the calibration chart shall have the same serial number as the modulator.

b. Transmitter, Radiosonde T-652( )/AMT-12.

c. Chart, Radiosonde Hypsometer, Calibration.

3.4.15.4 Identification Notice. - Each modulator shall have affixed to the battery compartment door, on the same side as the door catch and located centrally, an identification notice mounted so that the top of the notice i nearest the door catch. This notice shall have the following text:

"This instrument is a Radiosonde. It was used by the U.S. Armed Force: to obtain high-altitude weather data. It is not dangerous and is of n further use to the Government. Do not return. Dispose of it in any convenient manner."

3.4.16 Identification of Ground Check Test Leads. - Three test leads for purpose of baselining the radiosonde set shall be provided as follows:

a. Blue - low reference lead.

b. Yellow - humidity test lead.

c. Black - ground test lead.

The test leads in a, b and c above shall be brought through a single rainproof opening in the side of the case and shall extend 7 ±½ inches from the opening. The test leads shall be stripped of insulation at the end for ½ inch ±1/8 of an inch. (See 4.7.5)

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3.5 Weight. - Radiosonde Set AN/AMT-12( ), less batteries, shall weigh not more than 700 grams. (See 4.7.1)

3.6 Operating Voltages. - The radiosonde shall operate from a d.c. power source which supplies the following voltages measured at the transmitter connector:

Maximum	Minimum	Nominal
E_{a1}=7.4	E_{a1}=5.5	6.0
E_{a2}=1.5	E_{a2}=1.10	1.5
E_b=125	E_b=95	115

Where:

E_{a1} - R.F. Oscillator Filament

E_{a2} - Blocking Oscillator Filament

E_b = Common B+

3.7 Reflectivity. - All molded plastic case components shall have a reflectivity to solar radiation of 85 percent, in the visible light region, when compared to a freshly scraped magnesium carbonate block, or equal. (See 4.8.12)

3.8 Transmitter (T-652( )/AMT-12). - The transmitter shall transmit on a carrier frequency between 1660 MHz and 1700 MHz, which is 100% amplitude modulated (AM) by the output of the blocking oscillator at a rate determined by the meteorological data furnished by the modulator unit, and the reference resistor.

3.8.1 R.F. Oscillator Frequency. - The R.F. oscillator shall be tunable over the frequency range 1660 MHz to 1700 MHz. Operation of the R.F. oscillator shall be stable to within $\pm$6 MHz of any frequency in the above range. The R.F. oscillator shall be set at 1680 $\pm$4 MHz prior to delivery to the Government. (See 4.8.1)

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3.8.2 Super-regeneration. - The transmitter shall not super-regenerate at any time.

3.8.3 Pulse-width. - The rectangular pulse which modulates the R.F. oscillator shall blank the carrier for a duration of 45 to 125 microseconds. (See 4.8.3)

3.8.4 Modulation. - The R.F. oscillator output shall be 100 percent amplitude modulated by the blocking oscillator. (See 4.8.4)

3.8.5 Radiated Power Minimum. - The performance of the R.F. oscillator over the frequency range of 1660 MHz to 1700 MHz shall be such that after a stabilization period of 5 minutes at existing atmospheric conditions and with minimum voltages, the power output from the antenna shall be not less than 180 milliwatts. (See 4.8.5)

3.8.6 Plate Current. - With normal operating voltages, the plate current of the R.F. oscillator shall be not greater than 34 milliamperes. This shall be accomplished by proper selection of the grid resistor. (See 4.8.6)
NOTE: This selection is feasible within the tolerances specified for the grid resistors in 3.4.11.

3.8.7 Blocking Oscillator Grid Voltage or Current. - The blocking oscillator d.c. grid voltage shall not exceed -16 volts. Measurement of grid current may be substituted for measurement of grid voltage. The d.c. grid current shall not exceed 140 microamperes. (See 4.8.7)

3.8.8 Blocking Rate (Blocking Oscillator Leakage Resistance). - With the blocking oscillator grid circuit open, and the transmitter operating at normal voltages, the leakage resistance shall be such as to cause a blocking rate of not more than 15 pulses in any 60 second period under ambient atmospheric conditions. (See 4.8.8)

3.8.9 Blocking Oscillator Frequency. -

3.8.9.1 Reference Frequency. - With the grid resistor (112,400 ohms) externally grounded, the blocking oscillator frequency shall be 190 ±15 Hz. (See 4.8.9.1)

3.8.9.2 4/5 Reference Frequency. - With grid resistor (112,400 ohms) externally grounded through a resistance of 28,700 ± 1000 ohms, the blocking oscillator frequency shall be 4/5 of the reference frequency. (See 4.8.9.2)

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3.8.9.3 1/5 Reference Frequency. - With the grid resistor (112,400 ohms) externally grounded through a resistance of 457,000 + 8000 ohms, the blocking oscillator frequency shall be 1/5 of the reference frequency. (See 4.8.9.3)

3.8.10 Blocking Oscillator Frequency Stability. -

3.8.10.1 Blocking Oscillator Stability with Reduced Voltage. - The change in oscillator frequency, when voltages as specified in 3.6 are gradually reduced individually from maximum to minimum, shall not exceed 6 Hz. (See 4.8.10.1)

3.8.10.2 Blocking Oscillator Stability Due to Lead Dress. - The change in blocking rate of the blocking oscillator shall not exceed 1 Hz as determined in 4.8.10.2a and b. The lead dress effect as determined in 4.8.12c shall not exceed 1 Hz. (See 4.8.10.2)

3.8.10.3 Blocking Oscillator Stability with Voltage and Temperature Changes. - The change in blocking oscillator frequency at normal voltage shall not exceed 4 Hz for a temperature change from +60 C to +25 C, and shall not exceed 8 Hz between any two successive 15 minute interval measurements when tested in accordance with 4.8.12 (+25 C to -50 C) with simultaneous reduction of voltage from normal to minimum. (See 4.8.10.3)

3.8.10.4 Blocking Rate in High Humidity (Leakage Resistance). - Under the conditions specified in 4.8.13 the open circuit blocking rate shall not exceed 15 pulses every 60 seconds. (See 4.8.10.4)

3.9 Modulator MD-317( )/AMT-12. - The modulator provides the means for mounting the sensing elements (Temperature and Humidity), housing the battery, and contains the baroswitch assembly, switching circuitry, a hypsometer with a second relay for switching of the hypsometer circuit, and part of the input circuit to the blocking oscillator located in the transmitter.

3.9.1 Sensing Element Mounting Clips. -

3.9.1.1 Temperature Element Clips. -

3.9.1.1.1 Temperature Element Disengagement Force. - The temperature element clips shall accommodate the pig-tail leads of a Temperature Element, Resistance ML-419( )AMT-12 such that the leads shall not disengage when subjected to an axial pull of not less than 175 gms. (See 4.7.8)

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3.9.1.1.2 Leakage Resistance (Temperature Element Supporting Arm). - When subjected to the test of 4.8.11 the leakage resistance between the temperature element clips and the lower end of the temperature element supporting arms shall not be less than 1000 megohms.

3.9.1.2 Humidity Element Clips. - The humidity element clips of the modulator shall accommodate a Humidity Element, Resistance ML-476( )/AMT-12. The humidity element clips shall grip the humidity element securely, making good mechanical and electrical contact without causing physical damage to the element or the electrodes. (See 4.7.9)

3.9.1.3 Hypsometer Assembly. - The hypsometer, per SC-A-46884, is part of the MD-317( )/AMT-12, and the unit shall indicate pressure as a function of the boiling temperature of Trichlorofluoromethane (Freon 11). The hypsometer assembly consists of the following:

a. Dewar flask.

b. Thermistor.

c. Resistor.

d. Cap.

e. Wicking.

f. Support.

g. Calibration chart.

3.9.1.3.1 Dewar Flask. - The Dewar Flask shall have the following characteristics:

a. Capacity - 12 ±1 cubic centimeters.

b. Inside diameter - 15 ±2 millimeters.

c. Outside diameter - 26 ±2 millimeters.

d. Height - 85 ±5 millimeters.

e. Thermal loss - Less than or equal to 4°C (See SC-A-46884).

The flask shall be comparable to that manufactured by the American Thermos Bottle Company of Norwich, Connecticut or equal. (ATB Co Dwg No. A-10,721B)

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3.9.1.3.2 Thermistor. - The thermistor shall be a glass-stemmed bead type thermistor such as Western Electric Type 14B or VECO (Victory Engineering Co.) Type 32A1A, or equal.

3.9.1.3.3 Resistor. - An 820 ohm ±10% resistor shall be supplied with the hypsometer. The resistor wired to the 6 volt d.c. source shall be set within the flask near the base.

3.9.1.3.4 Cap. - The cap shall be manufactured of polyethylene per Specification MIL-P-22748, Class A, Grade 3 or other materials that have the same salient characteristics. The cap shall fit snugly over the Dewar Flask. An opening shall be provided in the cap for filling the hypsometer. The thermistor and resistor leads shall pass through smaller openings near the center of the cap.

3.9.1.3.5 Wicking. - The bead thermistor shall be supported such that the bead will be at or above the surface of the fluid (Freon 11) when the pressure is 50 mb or less. The bead shall be kept wet by means of wicking. The wick shall be of chemically clean muslin or bleached cotton woven into a 3/16 inch diameter tube. Approximately 4½ inches shall be used, with one end attached above the thermistor.

3.9.1.3.6 Support. - The hypsometer assembly shall be securely fixed in a vertical position in the modulator case. The supporting material will also function as a shock mount for the Dewar Flask.

3.9.1.3.7 Hypsometer Calibration Chart. - One hypsometer calibration chart will be supplied with each hypsometer. The chart will be a semi-logarithmic graph with the 50 to 1.5 millibars pressure range on the log ordinate and the linear abscissa representing the recorder division of the AN/TMQ-5 meteorological recorder paper. Each hypsometer calibration chart shall be stamped with the serial number of the radiosonde with which it is to be used.

3.9.1.3.8 Hypsometer Range and Accuracy. - The hypsometer shall operate over the pressure range between 50 and 1.5 millibars within the tolerance limits and under the service conditions specified herein. The accuracy limits shall be as follows:

Pressure Level	Tolerance
50 to 40 millibars	±1.0 millibar
40 to 20 millibars	±.5 millibars
20 to 1.5 millibars	±.25 millibars

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3.9.2 Baroswitch Assembly. - The baroswitch shall operate over the pressure range between 1060 millibars and 5 millibars (mb) within the tolerance limits and under the service conditions specified herein.

3.9.2.1 Linkage Mechanism. - The pressure capsule diaphragm, push pin and contact arm shall be properly aligned so as to minimize friction. The mechanism shall be effectively spring-loaded so as to reduce backlash to an absolute minimum. The mechanism shall operate properly at any temperature at which the modulator will be used. The entire mechanism shall be constructed so as to preclude any disengagement of any of the components thereof. (See 4.7.10)

3.9.2.2 Baroswitch Deflection. - The baroswitch shall have essentially a logarithmic relationship of deflection versus pressure such that its deflection per millibar in the region of 50 mb is at least four times but not more than seven times, the deflection in the region of 1,000 mb. The push pin shall not disengage from the pressure capsule when the arm is moved to the 150th contact at sea level pressure. (See 4.8.14.1)

3.9.2.3 Baroswitch Calibration Chart. - A baroswitch calibration chart showing atmospheric pressure in millibars versus contacts, obtained through the relay of the radiosonde at the "off" to "on" switching points, under conditions of decreasing pressure, for all contacts, shall be provided. The chart shall consist of two graduated scales printed adjacent to each other; one a linear scale reading pressure in millibars, the other reading contacts. The graduations on the contact scale shall be such that the initial contact shall represent a pressure not less than 1060 mb and the final contact a pressure not greater than 5 mb; distributed over at least 145 contacts. The graduations on the contact scale shall be marked with permanent ink or indelible pencil. The chart scale shall conform to SC-D-30164. Each calibration chart shall be stamped with the serial number of the baroswitch frame and with the serial number of the radiosonde modulator in which the baroswitch frame is contained and shall be legible in all respects. The calibration chart scale shall reflect the calibration of the instrument with which it is packed. Any irregularity of spacing between successive contacts shall not exceed 2 millibars for pressures greater than 350 millibars nor exceed 1 millibar for pressures less than 350 millibars. (See 4.7.13)

3.9.2.4 Setting Error. - The setting error is defined as the difference between the true pressure and the pressure indicated by the baroswitch, as obtained from the pressure calibration chart, at the beginning of the fifth contact, when the pressure is decreasing. This difference is considered a "positive" error when the indicated pressure is higher than the true pressure and "negative" when less than the true pressure. This setting error shall not exceed ± 4 millibars. (See 4.8.14.2)

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3.9.2.5 Calibration Accuracy. - The baroswitch, at 27 ±7 Centigrade, and after algebraic corrections for the setting error, shall provide an accuracy of ±4 millibars at pressures in excess of 150 mb, and an accuracy of ±2 mb at pressures of 150 mb or less. NOTE: Corrected chart pressure equals the indicated chart pressure minus the setting error times $\frac{A}{B}$.

Where A = pressure interval at any given contact.

And B = pressure interval of 5th contact.

The pressure interval is defined as the width of the contact in millibars. (See 4.8.14.3)

3.9.2.6 Low Temperature Effect. - When the baroswitch is subjected to a temperature which is 80 ±3 C below the room temperature run of 3.9.2.5 its calibration run pressure shall agree with the room temperature calibration to within ±4 mb at pressures in excess of 150 mb, and to within ±2 mb at pressures of 150 mb or less. (See 4.8.14.4)

3.9.2.7 Friction. - The deviation in observed pressure between the make point of a conducting segment under decreasing pressure conditions and the break point of the same conducting segment when the pressure is increasing shall not exceed the limits specified in a and b below. The rate of change of pressure shall be maintained at 100 ±5 mb/minute during the test. The unit shall be subjected to the full operating range of pressure change first decreasing then increasing the pressure at the rate specified above. (See 4.8.14.5)

	RANGE	MAX DEVIATION
a.	For pressure in excess of 150 mb	10 mb
b.	For pressure less than 150 mb	6 mb

3.9.2.8 Detent Operation. - The adjustment of the commutator in relation to the contact point shall have a range of not less than 100 mb, at sea level pressure. The commutator adjustment shall be accomplished by means of a detent contact screw such that each "click" of the adjustment knob shall move the commutator, relative to the contact point, a distance equivalent to a 0.5 ±0.1 mb pressure change, in the pressure range 1000 - 950 mb. Repeatability of the detent adjustment shall be within ±1 "click". (See 4.8.14.6)

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3.9.2.9 Bearing Tip. - The bearing tip of the contact point shall be polished and formed to provide a contact width of 0.003" max. and 0.001" min. (See 4.7.12)

3.9.2.10 Operational Width. - The combination of "contact width" of the contact point, the width of the commutator segments and the differential friction characteristics between conduction and insulating segments, when tested in the pressure region 1060 - 900 mb shall provide an operational width of 0.004 to 0.007 inch for the intermediate conducting segments. (See 4.8.14.7)

3.9.2.11 Contact Pressure. - The contact point shall remain in contact with the commutator when a lifting force of one gram is exerted at the tip of the contact arm immediately adjacent to the contact. Contact between the contact point and the commutator shall be broken when a lifting force of 2 grams is applied to the tip of the contact arm immediately adjacent to the contact point. (See 4.7.11)

3.9.2.12 Contact Resistance. - The total baroswitch and switching relay contact resistance shall not exceed 50 ohms when tested in accordance with 4.8.13.

3.9.2.13 Contact Arm Support. - The contact arm support of the baroswitch shall hold the bearing tip off the commutator from contact 0 to not less than contact 65 during shipment. The operation of the support shall be positive in both positions so that the arm will not be raised from the commutator during operation. (See 4.7.10j)

3.9.2.14 Commutator. - The commutator shall consist of 150 conducting segments each separated by an insulating segment. The surface of the commutator shall be polished smooth. The conducting segment shall be made of coin silver, gold plated in accordance with MIL-G-45204, Type II, Class I and paragraph 6.6(c) for under coat. The insulating segments shall be fabricated from a suitable plastic molding material (thermoplastic or thermosetting plastic) capable of meeting all specification requirements when exposed to all service conditions, per 3.10 and subject to approval of the Government. (See 4.7.7.1)

3.9.2.14.1 Adherence of Conducting Segments. - The adherence of the conducting segments to the plastic material shall be positive. (See 4.8.14.8)

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3.9.2.14.2 Commutator Segment Width and Thickness. - The physical width of the segments of the assembled commutator shall be as follows:

Each intermediate conducting segment: 0.004 - 0.005 inch

Each 5th conducting segment: 0.0065 - 0.0075 inch

Each insulating segment: 0.008 - 0.009 inch

The overall thickness of the commutator shall be 0.1875 ±0.010 inches. (See 4.7.7.2)

3.9.2.14.3 Leakage Resistance, Commutator. - The commutator leakage resistance, after cleaning and polishing the commutator, between metallic ground and all conducting segments of the commutator connected together, shall not be less than 100 megohms. The leakage resistance between any set of conducting segments shall meet the same requirements under the same conditions. (See 4.8.14.9)

3.9.2.14.4 Commutator Circuits and Associated Segments. - (See 4.7.7.3)

a. The humidity circuit will be activated when the contact point rests on the following commutator segments: 1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24, 26, 27, 28, 29, 31, 32, 33, 34, 36, 37, 38, 39, 41, 42, 43, 44, 46, 47, 48, 49, 51, 52, 53, 54, 56, 57, 58, 59, 61, 62, 63, 64, 66, 67, 68, 69, 71, 72, 73, 74, 76, 77, 78, 79, 81, 82, 83, 84, 86, 87, 88, 89, 91, 92, 93, 94, 96, 97, 98, 99, 101, 102, 103, 104.

b. The low reference circuits will be activated when the contact point touches the following contacts: 5, 10, 15, 20, 25, 35, 40, 50, 55, 65, 70, 80, 85, 95, 100, 106, 107, 108, 109, 111, 112, 113, 114, 116, 117, 118, 119, 121, 122, 123, 124, 126, 127, 128, 129, 131, 132, 133, 134, 136, 137, 138, 139, 141, 142, 143, 144, 146, 147, 148, 149.

c. The high reference circuits will be activated when the contact point touches the following contacts: 30, 45, 60, 75, 90, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150.

d. The hypsometer circuit will be activated when the contact point rests on the following commutator segments: 106, 109, 111, 114, 116, 119, 121, 124, 126, 129, 131, 134, 136, 139, 141, 144, 146, 148, 149.

e. The temperature circuits will be activated when the contact point touches the insulated portion between contacts of the commutator.

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3.10.1 Temperature. - During the test specified in 4.9.1 the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.8.1	R.F. Oscillator Frequency	Tunable over the range 1600 - 1700 MHz
3.8.3	Block, Osc. Pulse Width	45 - 125 microseconds
3.8.4	Modulation	100% AM
3.8.9.1	Reference Frequency	190 Hz to ±15 Hz
3.8.9.2	4/5 Reference Frequency	4/5 Meas Ref. Freq
3.8.9.3	1/5 Reference Frequency	1/5 Meas Ref. Freq

Before and after testing the equipment shall meet full specification performance for these requirements. In addition, requirement 3.9.2.14.1 must be met before and after testing. (See 4.8.14.8 and 4.9.1)

3.10.2 Humidity (moisture-resistance). - During testing as specified in 4.9.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.8.1	R.F. Oscillator Frequency	Tunable over the range 1660 - 1700 MHz
3.8.3	Block Osc. Pulse Width	45 - 125 microseconds
3.8.4	Modulation	100% AM
3.8.9.1	Reference Frequency	190 ±15 Hz
3.8.9.2	4/5 Reference Frequency	4/5 of Meas Ref Freq
3.8.9.3	1/5 Reference Frequency	1/5 of Meas Ref Freq
3.8.10.4	Blocking Rate in High Hum. (Leakage Resistance)	Not more than 15 pulses in any 60 sec period
3.9.1.1.2	Leakage Res of Temp Element Support Arm	Not less than 1000 megohms
3.9.2.14.3	Leakage Resistance, Commutator	Not less than 100 megohms

Before and after testing the equipment shall meet full specification performance for these requirements. In addition, the requirement of 3.9.2.14.1 must be met before and after testing. (See 4.8.14.8 and 4.9.2)

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3.10.3 Altitude (elevation). - The equipment shall meet full specification performance for the following measurements during testing per 4.9.3:

Paragraph	Measurement	Performance
3.8.1	R.F. Oscillator Frequency	Tunable over the range of 1660 - 1700 MHz
3.8.3	Block Osc. Pulse Width	45 - 125 microseconds
3.8.4	Modulation	100% AM
3.8.9.1	Reference Frequency	190 ±15 Hz
3.8.9.2	4/5 Reference Frequency	4/5 of Meas Ref. Freq
3.8.9.3	1/5 Reference Frequency	1/5 of Meas Ref. Freq
3.9.1.3	Calibration Accuracy of Hypsometer	50 to 40 mb ±1.0 mb 40 to 20 mb ±.5 mb 20 to 1.5 mb ±.25 mb
3.9.2.5	Calibration Accuracy of Baroswitch	±4 mb above 150 mb ±2 mb below 150 mb

After testing the equipment shall meet full specification performance for these requirements except para. 3.9.2.5.

3.10.4 Rain. - During the test specified in 4.9.4 the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement	Performance
3.8.10.4	Blocking Rate in High Hum. (Leakage Resistance)	15 pulses/15 sec.

After testing the modulator case and the transmitter case may be dried by wiping with a towel or rag. There shall be no evidence of water within the cases.

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3.10.5 Vibration. - The equipment shall be free from damaging resonances between 10 and 30 Hz during testing as specified in 4.9.5. The amplitude of vibration of the mounted equipment, including amplitude of parts and subassemblies shall not exceed twice the amplitude of the applied vibration during the test. After testing, the equipment shall meet the requirements of 3.8.5, 3.8.9.1, 3.8.9.2, 3.8.9.3, 3.9.1.3 and 3.9.2.5.

3.10.6 Bounce, Loose Cargo. - There shall be no degradation of performance after testing as specified in 4.9.5, Loose Cargo Bounce Test. After testing, the equipment shall meet the requirements of 3,8.5, 3.8.9.1, 3.8.9.2, 3.8.9.3, 3.9.1.3 and 3.9.2.5.

3.10.7 Shock, Transit Drop. - There shall be no degradation of performance after testing as specified in 4.9.6, Transit Drop Test. After testing, the equipment shall meet the requirements of 3.8.5, 3.8.9.1, 3.8.9.2, 3.8.9.3, 3.9.1.3 and 3.9.2.5.

3.11 Flight Test. - The radiosonde set shall perform satisfactorily both electrically and mechanically, as determined from the flight test record, and shall indicate temperature, pressure and relative humidity within the accuracy limits specified herein, when subjected to an actual flight test. (See 4.10)

3.12 Interchangeability. - Like units, assemblies and subassemblies and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.11)

3.13 Technical Literature. - Technical literature shall be furnished if specified in the contract.

3.14 Workmanship. - The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and applicable portions of the following paragraphs: (See 4.7)

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.

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Cleaning of units.
General requirement for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

b. In this specification:

3.4.3 Aneriod Capsule.
3.4.4 Baroswitch Frame and Support.
3.4.4.1 Hypsometer Assembly.
3.4.6 Case Assemblies.
3.4.7 Wire Hookup
3.4.8 Plugs and Receptacles.
3.4.13.1 Construction for Rainproofing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of ann inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - Inspection shall be classified as follows:

a. First Article Inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

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4.3 First Article Inspection. - This inspection shall be witnessed by the Government Commodity Engineering representative and shall consist of the inspection specified in subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B and Group C (See Tables IV, V and VI respectively). This inspection shall be performed in the order shown in Table III according to the number of samples specified in the contract or order.

TABLE III

ORDER OF INSPECTION

Inspection	Test No.	Number of Samples - 30 Units 1-3	Units 4-10	Units 11-15	Units 16-30	
Non Damaging						
Temperature	1	1	1	1	1	
Rain	2	2				
Altitude (elev)	3	3	3	3	3	
Vibration	4		4			
Potentially Damaging						
Humidity (Moist Resist)	6			6		
Fungus	7	7				
Shock; drop	8		8			
Flight Test	9					(12 add. units)

NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the non-damaging tests.

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4.4 Inspection Covered by Subsidiary Documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

ITEM	WHERE REQUIRED
Parts, Materials, Processes Used in Electronic Equipment	3.4
Relay	3.4.2
Wire, Hookup	3.4.7
Electron Tubes	3.4.9
Resistors	3.4.11.2
Temperature Element ML-419( )/AMT	3.4.12
Humidity Element ML-476( )/AMT	3.4.12
Marking	3.4.15
Polyurethane	3.9.1.3.4

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. - The contractor shall perform the inspection specified in 4.4 through 4.5.4, except for the flight test which will be performed solely by the Government. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table IV. Each lot shall be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table IV. Group A inspection shall be performed in any order which is satisfactory to the Government, except that operational inspection shall be last.

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TABLE IV - GROUP A INSPECTION

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual Mechanical	3.14	4.7	1.5%	6.5%
Ancillary Items	3.4.12	4.7.6	1.5%	6.5%
Ground Check Test Leads	3.4.16	4.7.5	1.5%	6.5%
Operating Voltages	3.6			
Frequency of R.F. Oscillator	3.8.1	4.8.1	1.0%	
Super-regeneration	3.8.2	4.8.2	1.0%	
Pulse Width	3.8.3	4.8.3	1.0%	
Modulation	3.8.4	4.8.4		
Radiated Power (Minimum)	3.8.5	4.8.5		
Plate Current	3.8.6	4.8.6		
Oscillator Grid Voltage or Current	3.8.7	4.8.7		
Reference Frequency	3.8.9.1	4.8.9.1		
4/5 Reference Frequency	3.8.9.2	4.8.9.2		
1/5 Reference Frequency	3.8.9.3	4.8.9.3		
Blocking Oscillator Stability with Reduced Voltage	3.8.10.1	4.8.10.1		
Hypsometer Calibration Chart	3.9.1.3.7	4.8.14.10		
Hypsometer Range and Accuracy	3.9.1.3.8	4.8.14.10		
Baroswitch Deflection	3.9.2.2.	4.8.14.1		
Baroswitch Calibration Chart	3.9.2.3	4.7.13		

1.0% for the group

(continued on next page)

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Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Setting Error	3.9.2.4	4.8.14.2	(continued from preceding page)	
Calibration Accuracy of Baroswitch	3.9.2.5	4.8.14.3		
Friction	3.9.2.7	4.8.14.5		
Contact Pressure (Baroswitch)	3.9.2.11	4.7.11	1.5%	6.5%
Contact Resistance	3.9.2.12	4.8.13	1.0%	
Contact Arm Support	3.9.2.13	4.7.10	1.5%	6.5%

4.5.2 Group B Inspection. - This inspection, including sampling shall conform to Table V and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection. The inspection level shall be S-4 for normal, tightened, and reduced inspection, and the AQL shall be 6.5%.

TABLE V - GROUP B INSPECTION

Inspection	Req Para	Insp Para	AQL
Resistors	3.4.11.1 & 3.4.11.2	4.7.14	6.5% combined for the group
Temperature Element Clips	3.9.1.1.1	4.7.8	
Humidity Element Clips	3.9.1.2	4.7.9	
Low Temperature Effect	3.9.2.6	4.8.14.4	
Bearing Tip	3.9.2.9	4.7.12	
Baroswitch Contact Point Operational Width	3.9.2.10	4.8.14.7	
Commutator Circuits and Associated Segments	3.9.2.14.4	4.7.7.3	
Interchangeability	3.12	4.11	

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4.5.3 Group C Inspection. - This inspection shall consist of the tests specified in Table VI and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Each sample unit used for the humidity or fungus test shall be considered destroyed at the completion of testing and shall not be reconditioned or repaired (See 4.5.4) for acceptance on contract. Sample units shall be selected in accordance with paragraph 4.5.3.1.

TABLE VI - GROUP C INSPECTION

Inspection	Req Para	Insp Para
Group 1		
Flight (Note 1)	3.11	4.10
Group 2		
Fungus (Note 2)	3.4.14	4.9.7
Detent Operation	3.9.2.8	4.18.14.6
Temperature	3.10.1	4.9.1
Humidity (Note 2)	3.10.2	4.9.2
Elevation	3.10.3	4.9.3
Rain	3.10.4	4.9.4
Group 3		
Blocking Oscillator Stability Due To Lead Dress	3.8.10.2	4.8.10.2
Blocking Oscillator Stability With Temperature Changes	3.8.10.3	4.8.10.3
Blocking Rate In High Humidity	3.8.10.4	4.8.10.4

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Group 4

Adherence of Commutator Segments	3.9.2.14.1	4.8.14.8
Commutator Segment Width and Thickness	3.9.2.14.2	4.7.7.2
Leakage Resistance:		
Oscillator Blocking Rate	3.8.8	4.8.8
Temperature Element Supporting Arm	3.9.1.1.2	4.8.11
Baroswitch Commutator	3.9.2.14.3	4.8.14.9

Group 5

Weight	3.5	4.7.1
Case Reflectivity	3.7	4.8.12
Vibration and Bounce	3.10.5 & 3.10.6	4.9.5
Shock; Drop	3.10.7	4.9.6

NOTE 1: Units subjected to Sub-Group 1 inspection are considered destroyed and are in addition to specified contractual delivery quantities.

NOTE 2: Units subjected to these tests are considered destroyed and shall not be included in shipment lots.

4.5.3.1 Sampling for Inspection. -

a. First Article (Preproduction). - When First Article samples are required by the contract or purchase order, the number of samples specified shall be fabricated and shall be subjected to Group A, Group B and Group C inspections. Environmental testing shall conform to Table III.

b. Initial Production. - When Initial Production samples are specified in the contract or purchase order, thirty (30) each Radiosonde Set (complete) shall be randomly selected, by the Government, from the first production lot shipped and subjected to Group A, Group B and Group C inspections. Environmental testing shall conform to Table III.

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c. Subsequent Production. - Subsequent to either First Article or Initial Production samples per a and b above, twelve (12) each Radiosonde Set (complete) shall be randomly selected, by the Government, from production shipment lots in each three (3) month period. These samples shall be selected from production lots which have been subjected to and passed Group A and Group B inspections in the following manner:

Group C Sub-Group No.	No of Samples
Group 2	3
Group 3	3
Group 4	3
Group 5	3

d. Flight Test. - Twelve (12) each Radiosonde Set (complete) shall be randomly selected by the Government, from each month's shipment and subjected to the inspection of Sub-Group 1, Group C. If there is a final lot it shall be treated as a full month's production.

4.5.3.2 Noncompliance. - The contractor shall immediately report, in writing, each in-plant Group C failure occurrence, including details of the failure and the characteristics affected. The Government shall promptly refer all Group C flight test failures to the contractor for investigation and correction, whereupon the contractor shall immediately investigate the cause of failure and further report the results of the investigation and details of corrective action on:

a. The processes and materials, as applicable.

b. All units of product which were manufactured under the same conditions and which the Government considers subject to the same failure.

Reports shall be forwarded to the responsible technical activity designated in the contract, through the Government Quality Assurance Representative (QAR). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspection(which the sample failed, at the option of the Government, and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

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4.5.4 Reinspection of Conforming Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection, except Sub-Group 1 of Table VI, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.6 Test Equipment and Test Conditions. - Test equipment shall meet the calibration requirements of MIL-C-45662. A complete list of test equipment including model and serial number to be used in First Article and Production Tests shall be submitted with the proposed First Article or Initial Production Test procedure. If a black box is to be used, a schematic and breakdown of all parts shall be submitted with the proposed test procedure.

4.6.1 Pressure Standard. - The contractor shall provide a standard of pressure measurement in the form of a mercurial barometer, for standardizing of the calibration and inspection pressure measuring equipment used in the manufacture of the pressure units. The barometer shall cover the full operating range of the modulator and shall provide measurements of absolute pressure to within 0.01 inch of mercury (0.34 millibar) after all corrections are applied. Copies of certificates of accuracy of the standard used, issued by a competent authority within one year prior to each time it is used on the contract, shall be furnished by the contractor to the Government inspector.

4.6.2 Maintenance of Accuracy. - The absolute accuracy of all required pressure measuring equipment used for the calibration and inspection of the pressure unit shall be maintained at all times to within 0.5 mb of the standard. The measuring equipment used shall be calibrated against the standard specified in 4.6.1, before the beginning of the inspection, and at regular monthly intervals thereafter.

4.6.3 Temperature. - Temperature measuring equipment with an accuracy of measurement of ± 1 C in the range of ± 90 C shall be provided for use in the temperature tests specified herein.

4.6.4 Resistance. - A wheatstone bridge with an accuracy within ± 1 percent in the range of 0 to 5000 ohms, and a megohmeter with an accuracy within ± 2 percent in the range of 0.5 to 50 megohms and within ± 10 percent in the range of 50 to 3000 megohms shall be provided by the contractor for use in the tests specified herein.

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4.6.5 Pressure Chambers. - Chambers shall be provided for checking the pressure units at pressures over the range of 1060 mb to 2 mb at a temperature of 25 ±5 C. Connection to the test chamber for the pressure measuring equipment shall be made with sufficiently large tubing to eliminate any pressure gradients between the test chamber and the measurement equipment. Electrical connections for measurement of required test data in the radiosonde shall be provided in the chamber.

4.6.6 Temperature Pressure Chamber. - A chamber suitable for testing the instruments over the pressure range of 1060 mb to 10 mb and maintaining simultaneously a temperature as low as -60 ±2 C shall be provided for running tests to determine temperature effects on the radiosonde. (Pressure connections for the pressure measuring equipment for this chamber shall be as specified for the test chamber described in 4.6.5.)

4.6.7 Radiosonde Blocking Oscillator and Frequency Measuring Equipment. - The radiosonde blocking oscillator frequency measuring and associated test equipment used shall be checked once every week in the following manner:

a. Insert the resistance noted below (corresponding to representive selected temperatures) in the grid circuit of the blocking oscillator, at the point normally occupied by the temperature element. Obtain a reference before and after each reading and apply corrections, if necessary.

+50 C=24,200 ohms	±0.1%
+30 C=40,000 ohms	±0.1%
0 C=94,040 ohms	±0.1%
-30 C=262,400 ohms	±0/1%
-60 C=928,000 ohms	±0.1%

b. Using the data obtained above and Temperature-Humidity Evaluator CP-223/UM, per MIL-C-55076 (lock-in at 30 C), evaluate the frequencies obtained in terms of temperature (Government Loaned Equipment).

c. The difference between the true temperature and the evaluated temperature for each of the points shall not be greater than 0.25 C.

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4.6.8 Vibration. - Except where a specific vibration test is listed all of the tests on the radiosonde set shall be so conducted that there will be no vibration transmitted to the instrument under test. This requirement applies particularly to the calibration, friction, and cold-temperature tests on the pressure unit.

4.6.9 Temperature. - Unless other temperature conditions are specified, all tests on the radiosonde set shall be made at 27 ±7 C. A specific temperature shall be maintained for a period of 15 minutes prior to measuring and recording test data. In tests at a fixed temperature, the temperature of the base of the modulator shall be considered as the test temperature.

4.6.10 Pressure. - In the detailed calibration and friction test, and the calibration at room and low temperature tests of the pressure unit, the pressure in the test chamber shall be reduced at as fast a rate as possible except while making observations, when the rate of change of pressure shall not exceed 1 mb every 5 seconds. In all pressure tests the pressure in the chamber shall be reduced or increased uniformly and without reversal.

4.7 Visual and Mechanical Inspection. - The equipment shall be examined to determine conformance to the applicable requirements listed in MIL-STD-252 and Table VII below and in accordance with the following subparagraphs.

TABLE VII - Classification of Visual and Mechanical Defects

Classification	Defect
Major	Contact arm support does not lock positively in storage or operating position.
	Temperature element clips do not meet pull test.
	Rainproof sealant missing.
	Commutator segments too wide, too narrow.
	Connection not soldered; insecurely soldered.
	Commutator segments do not adhere to commutator surface.
	Commutator surface rough; uneven.
	Rivets, eyelets, mechanical fasteners used to make electrical connections; loose, connection intermittent; connection not made.

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Classification	Defect
	Insulation damaged, bare wire exposed, may cause short circuit.
	Humidity element clips damaged; do not make good electrical contact with element.
	Ground check test leads too short; connected to wrong point of termination.
	Baroswitch contact pressure less than 1 gm or more than 4 gm.
	Case assembly warped, covers do not fit; cannot be closed.
	Baroswitch loosely mounted.
	Hypsometer loosely mounted.
	Case assembly damaged; cracked.
	Antenna or cone reflector damaged; silver plating missing from coaxial tubing.
	Unit overweight.
	Baroswitch calibration chart wrong; not numbered.
	Hypsometer calibration chart wrong; not numbered.
	Baroswitch improperly assembled; affecting operation.
	Hypsometer improperly assembled; affecting operation.
Minor	Ground check test leads too long.
	Solder excessive; cold solder connection.
	Rainproof sealant inadequate.

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Classification	Defect
	Case assemblies distorted not affecting operation or closure.
	Marking missing, or illegible, improper.
	Insulation damaged, bare wire exposed, no chance of short circuit or cross.
	Baroswitch contact pressure exceeds 2 gms but is less than 4 gms.
	Cut or machined plastic parts; not smooth; molding flash not removed.
	Contact arm support does not support contact arm to minimum number of contacts.

4.7.1 Weight. - The weight of the radiosonde set shall be measured using a scale accurate to within 2% for compliance with 3.5.

4.7.2 Case Assembly. - The case assemblies shall be inspected to insure the following:

a. That all openings which would allow incursion of moisture during the rain test are adequately sealed.

b. That the case is not warped, is free from molding "flash" and other molding imperfections (cold flow, voids, etc.).

c. That the top cover and battery compartment doors close tightly and fasten securely.

d. That the temperature elements support arms are securely mounted, lock in place in the operating configuration, are not distorted or warped, and are free from molding imperfections.

e. That the humidity element rain shield fits properly into mounting holes, is not warped or distorted and locks securely in place in the operating configuration.

f. That the transmitter case assembly is not distorted, is free from molding flash or other molding imperfections.

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4.7.3 Wires and Plugs. - Wires and plugs shall be inspected for physical conformance to the model and 3.4.7 and 3.4.8.

4.7.4 Antenna System. - The antenna system shall be inspected for damage, deformity, and to determine that the coaxial tubing has been silver plated in accordance with paragraph 3.4.10.

4.7.5 Ground Check Test Leads. - The ground check test leads shall be inspected for conformance with 3.4.16.

4.7.6 Ancillary Items. - The ancillary items shall be inspected for conformance with 3.4.12 and 3.4.12.1.

4.7.7 Commutator. -

4.7.7.1 Commutator Contour. - The commutator shall be examined for smoothness and uniformity of surface. The edges of the conducting segments shall be sharp, clear, free from burrs. There shall be no evidence of peeling or bubbling of the conducting segments. The commutator shall conform to 3.9.2.14.

4.7.7.2 Commutator Segment Width and Thickness. - The width and thickness of commutator conducting segments shall be checked to determine compliance with 3.9.2.14.2 using a calibrated tool-makers microscope or equivalent.

4.7.7.3 Commutator Circuits and Associated Segments. - Commutator circuits and associated segments shall be inspected for compliance with 3.9.2.14.4.

4.7.8 Temperature Element Clip. - A temperature element pigtail lead or wire gauge of equivalent diameter shall be inserted into the temperature element clip. A pull type spring balance gauge with a mid-range reading of 175 gms and an accuracy of ±5% shall be connected to the lead or wire gauge close to the clip. The force shall be exerted in an axial direction which would most easily disengage the element lead from the clip. The force necessary for disengagement shall comply with 3.9.1.1.1.

4.7.9 Humidity Element Clips. - The humidity element clips shall be inspected using a humidity element or equivalent gauge block to determine compliance with 3.9.1.2.

4.7.9.1 Hypsometer Calibration Chart. - The hypsometer calibration chart shall be inspected for compliance with 3.9.1.3.7. Failure to meet any requirement of 3.9.1.3.7 constitutes a major defect.

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4.7.10 Baroswitch Subassembly. - The baroswitch subassembly shall be inspected for the following visual and mechanical attributes, with the baroswitch mounted in the radiosonde case:

a. Pivots and shaft bearings shall be checked to see that some end play exists and that there is no binding.

b. Connector pins and linkage mechanism shall be properly aligned with respect to the contact arm.

c. Connector pin bearings are free of burrs.

d. All required backlash springs are in place and that tension is applied to the linkage mechanism by the springs.

e. Lubricants, if any, used on any of the bearings of the linkage mechanism shall be checked for suitability at low temperature extremes (-90 C).

f. Welding or soldering of flat springs or suspension members shall be checked for excessive soldering flux, and evidence of over-heated or under-heated soldered or welded joints.

g. Contact arm bearing or suspension shall be visually inspected for alignment with the commutator surface.

h. The detent adjusting screw shall be checked to see that it neither binds nor slips.

i. The contact arm support shall be inspected to determine that its action is positive in both storage and operating position.

j. With the contact arm support in the storage position the contact arm shall be gently moved, manually, to determine that the contact point does not touch the commutator before the 65th contact segment.

4.7.11 Contact Point Pressure. - The contact point pressure shall be inspected for compliance with 3.9.2.11.

4.7.12 Bearing Tip. - The contact point bearing tip shall be inspected with a shadowgraph or calibrated toolmakers microscope for compliance with 3.4.1 and 3.9.2.9.

4.7.13 Baroswitch Calibration Chart. - The baroswitch calibration chart shall be inspected for compliance with 3.9.2.3.

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4.7.14 Resistors. - Resistors shall be measured with a Wheatstone Bridge or equivalent for compliance with 3.4.11.1 and 3.4.11.2.

4.8 Performance Tests. - The equipment unit complete shall be subjected to the following inspections under the test conditions specified.

4.8.1 R.F. Frequency. - The R.F. oscillator frequency shall be measured, using suitable frequency measuring test equipment as follows:

a. Operating voltages shall be adjusted to the maximum values listed in 3.6.

b. The operating voltages shall then be gradually varied individually from their respective maximum to minimum values listed in 3.6.

c. At no time during this test shall the deviation of R.F. frequency exceed the tolerance of 3.8.1.

4.8.2 Super regeneration. - The transmitter output shall be monitored using a spectrum analyzer to determine compliance with 3.8.2. During this test the supply voltages shall be varied as specified in 4.8.1a and b.

4.8.3 Pulse Width. - The transmitter shall be placed within a R.F. coupler having a VSWR of 1.4:1 when terminated in a 50 ohm impedance. The detected modulation pulse width displayed on a calibrated oscilloscope shall conform to the requirement of 3.8.3.

4.8.4 Modulation. - Percent modulation shall be determined by heterodyning the modulated 1660 MHz - 1700 MHz radiated signal with an intermediate frequency which is within the band-pass of the oscilloscope used. The resultant shall be viewed on the scope to determine that the amplitude of the displayed signal becomes zero, at the time of, and for the duration of each modulating pulse in compliance with 3.8.4.

4.8.5 Radiated Power (Minimum). -

a. Operate the transmitter at the minimum voltages of 3.6 for at least five (5) minutes; and

b. Place the transmitter operating at minimum specified voltages in the coupler of 4.8.3; then

c. Measure the radiated power output with an approved power meter, in accordance with 4.6, matched with a 3-stub tuner impedance matching system. The measured power shall comply with 3.8.5.

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4.8.6 Plate Current (R.F. Oscillator). - The R.F. oscillator plate current shall be measured with an approved milliammeter for compliance with 3.8.6.

4.8.7 Blocking Oscillator Grid Voltage or Current. - The blocking oscillator grid voltage or grid current shall be measured with either an approved voltmeter or microammeter respectively to determine compliance with 3.8.7.

4.8.8 Blocking Rate (Blocking Oscillator Leakage Resistance). - Leakage resistance shall be determined by checking the open-circuit frequency of the blocking oscillator with an electronic counter or other approved means for compliance with 3.8.8.

4.8.9 Blocking Oscillator Frequency Test Procedure. - With the radiosonde operating at normal voltages, per 3.6, and with the grid return circuit externally grounded through a precision decade resistance for the tests of 4.8.9.1 thru 4.8.9.3 an electronic counter or other approved means shall be used to measure the frequency of the blocking oscillator signal after it has been detected from the radiated transmitter output. The temperature shall be maintained at 25±5°C and the supply voltage shall be checked prior to each measurement.

4.8.9.1 Reference Frequency. - With the transmitter set up for test as in 4.8.9 the reference frequencies shall be checked for compliance with 3.8.9.1.

4.8.9.2 4/5 Reference Frequency. - The 4/5 reference frequency shall be checked by placing the precision decade resistance externally in the grid circuit of the blocking oscillator. The resistance shall be varied until the detected blocking oscillator frequency is exactly 4/5 of the reference frequency measured in 4.8.9.1. The resistance value at this point shall be recorded and shall be in compliance with the value specified in 3.8.9.2.

4.8.9.3 1/5 Reference Frequency. - The 1/5 reference frequency test shall be performed as in 4.8.9.2 except that the resistance shall be varied until the detected frequency is exactly 1/5 of the reference frequency measured in 4.8.9.1. The resistance value shall be recorded and shall comply with 3.8.9.3.

4.8.10 Blocking Oscillator Frequency Stability Tests. - The following tests shall determine compliance with the requirements of 3.8.10.

4.8.10.1 Reduced Voltage Effect on Blocking Oscillator Frequency. - With the transmitter connected to the electronic counter as stipulated in 4.8.9 the input voltages shall be gradually reduced from the maximum to the minimum as specified in 3.6. The change in oscillator frequency from maximum to minimum shall not exceed the requirements stipulated in 3.8.10.1.

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4.8.10.2 Lead Dress Effect on Stability of Blocking Oscillator. - This test shall be performed in the following manner:

a. With the blocking oscillator operating at reference frequency, decouple the blocking oscillator from the R.F. oscillator by disconnecting the 0.1 mfd capacitor. Adjust the 110 volt and 1.4 volt supplies to normal and leave the 6.0 volt filament supply open. Connect the output of the blocking oscillator to a suitable measuring device. All other operating conditions being normal, the blocking oscillator output shall comply with 3.8.10.2.

b. The 6.0 volt filament supply shall be connected to the R.F. oscillator, operating with normal B+ voltage. After the R.F. oscillator is operating for not less than one minute the output frequency of the blocking oscillator shall be noted.

c. With the R.F. oscillator operating and the 0.1 mfd capacitor reconnected, twist the transmitter grid and power leads until the blocking oscillator rate changes from the lowest to the highest frequency. The difference in output readings shall determine compliance with 3.8.10.2.

4.8.10.3 Blocking Oscillator Stability with Voltage and Temperature Changes.

a. Operate the radiosonde at normal voltage and room temperature for five (5) minutes and record blocking oscillator frequency; then

b. Rapidly raise the temperature to +60 C. Following a ten (10) minute stabilization period record the frequency; then

c. Reduce the temperature at a uniform rate to 25 C in two equal steps during a 20 minute period with normal voltages applied, record the frequency. The difference between b and c shall not exceed 4 Hz; then

d. Simultaneously reduce the temperature from +25 C to -50 C and the voltages from normal to minimum uniformly over 6 equal steps of 15 minutes duration, allow a ten (10) minute stabilization period at -50 C. The frequency shall be recorded at the end of each step immediately prior to the next successive reduction. The frequency shall remain stable within the limit specified in 3.8.10.3.

4.8.10.4 Blocking Rate and Reference Frequency in High Humidity (Leakage Resistance). -

a. Operate at normal voltage and room temperature for 5 minutes.

b. Measure and record the blocking rate of 4.8.8.

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c. Increase the temperature to 40 ±5°C, then raise the humidity to 90 ±5% and hold the radiosonde in this environment for a minimum of 15 minutes. Measure and record the reference frequency and the open circuit blocking rate for compliance with 3.8.10.4.

4.8.11 Leakage Resistance, Temperature Element Supporting Arm. - The radiosonde shall be subjected to a relative humidity of 90 ±2% at a temperature of 25 ±5°C for not less than two (2) hours. Immediately thereafter the leakage resistance shall be measured, using an approved megohmmeter, for compliance with 3.9.1.1.2.

4.8.12 Case Reflectivity. - Reflectivity of the case assemblies shall be measured in the visible light region using a General Electric Hardy recording spectrophotometer, or equal, as follows:

a. Using a freshly scraped magnesium carbonate block, or equal, record the reflectance across the visible light region of the solar spectrum.

b. Measure the reflectance of radiosonde case components in the same visible light region of the solar spectrum.

c. The graphic records of reflectance shall be compared and at no point shall the case sample graph show a variance greater than that specified in 3.7.

4.8.13 Contact Resistance. - With the baroswitch contact point centered on a conducting segment and with normal operating contact pressure, the contact resistance shall be measured between ground and the respective made terminals of the switching relay with the relay first in the operated, then in the unoperated mode. The measured contact resistance shall conform to 3.9.2.12.

4.8.14 Modulator Operation. - The tests of 4.8.14.1 thru 4.8.14.7 shall be conducted in a pressure chamber in which the pressure can be controlled between the limits of 1060 mb to 5 mb, and with temperatures controllable over the range of +27°C to -53°C. Facilities shall be provided for connecting the test equipment external to the chamber.

4.8.14.1 Baroswitch Deflection. - The modulator shall be placed in a pressure chamber per 4.6.5. The pressure shall be decreased uniformly from 1060 mb to 5 mb. A recording of contact make and break shall be made on a brush recorder or equal. The increment between the make of successive intermediate contacts shall be measured to determine compliance with 3.9.2.2.

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4.8.14.2 Setting Error Test. - The modulator shall be placed in the pressure chamber and the pressure reduced until the fifth contact just makes, at which time the true chamber pressure shall be noted. This reading shall be subtracted from the calibration chart reading for the fifth contact to give the setting error, as specified in 3.9.2.4.

4.8.14.3 Calibration Accuracy Test. - The modulator assembly with the contact arm lowered on the commutator shall be placed in the test chamber at room temperature (27 ±7°C) and electrical connections shall be made to indicating instruments external to the chamber. The pressure in the chamber shall be increased to a pressure between 1060 and 1030 mb and then reduced at the rate specified in 4.6.10. The true pressure in the chamber shall be observed at the make of the fifth contact, and at any contact on the pressure unit, through its range, with a final observation at a maximum pressure of 10 mb. (A minimum of 15 observations shall be made.) The pressure shown on the calibration chart of the modulator assembly, after corrections for setting error as described in 3.9.2.4 shall be as specified in 3.9.2.5.

a. At pressures in excess of 150 millibars deviations of more than 4 mb but not more than 7 mb constitute a minor defect; deviations of more than 7 mb constitute a major defect.

b. At pressures below 150 millibars deviations of more than 2 mb but not more than 4 mb constitute a minor defect; deviations of more than 4 mb constitute a major defect.

c. A setting error between 4 and 7 mb constitutes a minor defect; deviations of more than 7 mb constitute a major defect.

4.8.14.4 Low Temperature Effect. - After the test of 4.8.14.3 and without adjustment or resetting, the aneroid unit shall be tested at a temperature which is 80 ±3°C below that used for the test of 4.8.14.3. The unit shall be placed in the test chamber and brought to temperature equilibrium, after which the pressure shall be reduced and observations taken at each contact made in 4.10.2. The true pressure for each observed contact shall agree with the true pressure for the same contact in the test of 4.8.14.3 to within the limits specified in 3.9.2.6. Minor and major defects are defined in 4.8.14.3a thru c.

4.8.14.5 Friction. - The difference in true pressure for each 15th contact, observed at the make of the contact during decreasing pressure and the break for increasing pressure, shall comply with 3.9.2.7.

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4.8.14.6 Detent Operation. -

a. The pressure at the beginning of both the 10th and 11th contacts shall be determined from the baroswitch calibration chart.

b. Turn the detent adjusting screw until contact is just made with the leading edge of the 10th contact. Then turn the screw clockwise, counting the clicks, until contact is just made with the leading edge of the 11th contact.

c. The relative movement of the commutator, with respect to the contact points, for each click shall be computed by dividing the pressure differential of a by the number of clicks of b to determine compliance with 3.9.2.8.

d. The detent adjustment screw shall be turned clockwise for fifteen clicks and then turned counter-clockwise until contact is just made with the leading edge of the 11th contact. The difference between clockwise and counter-clockwise clicks shall be used to determine compliance with the repeatability of 3.9.2.8.

NOTE: Deviations constitute minor defects.

4.8.14.7 Operational Width. - From the observed pressure, the operational width in terms of pressure of each intermediate conducting and insulating segment shall be determined in the pressure region of 1060 - 900 mb. The pressure intervals shall then be translated into equivalent dimensions of the intermediate conducting segments to determine compliance with 3.9.2.10.

4.8.14.8 Adherence of Conducting Segments. - A strip of pressure sensitive tape, per L-T-90 shall be applied over the printed circuit portion of the commutator with thumb-nail pressure. The tape shall then be removed slowly (as differentiated from a quick pull). After removal of the tape there shall be no evidence of blistered or removed portions of the printed circuit. This test shall be performed and after the temperature test of 3.10.1 and the humidity test of 3.10.2.

4.8.14.9 Leakage Resistance (Commutator). - After exposure for two (2) hours to an atmosphere with a relative humidity of 90 ±2 percent at a temperature of 25 ±5°C, the leakage resistance shall be measured using an approved megohmmeter for compliance with 3.9.2.14.3.

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4.8.14.10 Hypsometer Calibration Accuracy. - The hypsometer shall be tested as follows:

a. Each hypsometer using approximately 5 cc of Freon 11 (Trichlorofluoromethane) shall be placed in a pressure calibration chamber.

b. The hypsometer shall be connected into the grid-to-ground circuit of the Transmitter, Radiosonde, T-652( )/AMT-12 (or equivalent).

c. Readings of frequency or recorder divisions on Radiosonde Recorder AN/TMQ-5( ), or equal, shall be obtained at the following pressures: 50, 20, 10, 6, 3 and 2 millibars and any additional points deemed necessary. A calibration curve shall be constructed using the readings obtained. This curve shall be compared to the original calibration curve and agree to it within the accuracies specified in 3.9.1.3.8.

4.9 Service Condition Tests. -

4.9.1 Temperature. - The radiosonde shall meet full specification performance for the measurements specified in 3.10.1 during the following tests:

a. High Temperature. - The equipment shall be subjected to the test of Method 501, Procedure II of MIL-STD-810. At Step 7, the internal temperature shall be adjusted to 71°C for operational tests (this includes the effects of solar radiation).

b. Low Temperature. - The equipment shall be subjected to the test of Method 502, Procedure I of MIL-STD-810, storage temperature, Step 2, shall be -90°C and shall be maintained for 3 hours after the chamber temperature has stabilized. Operating temperature, Step 4, shall be maintained at -90°C.

4.9.2 Humidity. - The radiosonde shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810, except that Step 6 shall be two (2) continuous 48 hour cycles. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.10.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not remove the equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. Any radiosonde which fails to meet the specified performance during cycling, shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If any equipment fails to meet specified performance during cycling after one readjustment or realignment, or fails subsequent to cycling, that equipment does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

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4.9.3 Elevation (Altitude) Test. - The equipment shall be put in a temperature altitude chamber and shall be tested according to the following procedures:

Step 1: The internal pressure, temperature and humidity of the test chamber shall be at standard sea level conditions. While maintaining these conditions, the equipment shall be operated and shall meet full specification performance for the measurements specified in 3.10.3.

Step 2: With the equipment still operating, the temperature and pressure shall be changed simultaneously from that at standard sea level conditions to those at 105,000 feet at the following rates:

a. Temperature:

Sea Level to 36,000 ft:	-2.0°C/min.
36,000 ft to 65,000 ft:	Constant at -57°C
65,000 ft to 105,000 ft:	+0.4°C/min from -57°C base

b. Pressure:

Equivalent to an altitude change of 1,000 ft/min.

Performance of 3.10.3 shall be recorded every 5 minutes during the test.

Step 3: After testing per Steps 1 and 2 the equipment shall meet full specification performance for the measurements specified in 3.10.3.

Step 4L Comparison of test data recorded in Steps 1, 2 and 3 shall indicate no deviation from specified performance characteristics beyond the limits specified herein.

4.9.4 Rain. - The radiosonde shall be subjected to the test of Method 506, Procedure I, of MIL-STD-810. During each 30 minute rain period, the wind shall be maintained for 15 minutes commencing 5 minutes after the start of the period. The equipment shall be operated during the last 10 minutes of each 30 minute period. The equipment shall meet full specification performance for the measurements specified in 3.10.4. In addition, an open circuit blocking rate greater than ½ Hz shall be considered a failure. Evidence of water in either the modulator or the transmitter shall be considered a failure.

4.9.5 Vibration and Bounce, Loose Cargo. - The equipment shall be subjected to the test of Method 514, Procedure XI, Parts 1 and 2 of MIL-STD-810.

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4.9.6 Shock Test; Transit Drop. - The equipment shall be tested in accordance with Method 516, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance.

4.9.7 Fungus. - The equipment shall be subjected to the test of Method 508, Procedure I of MIL-STD-810. Gasket seals or other nonhermetically sealed test items shall be opened during exposure and internal areas shall be sprayed with spore suspension as well as external surfaces. Operation of the equipment is not required, but examination at the conclusion of the exposure period shall reveal no evidence of viable fungus on any surface.

4.10 Flight Test. - The completely assembled radiosonde, including parachute and balloon shall be released in actual flight. The transmitted signals shall be received by the ground station equipment specified in 3.1 or Government approved equivalent. The data shall be recorded on a Weather Data Recorder AN/TMQ-5( ) or Government approved equivalent and shall conform to 3.11. This test will be conducted by the Government at a site selected by the Government, and with receiving equipment furnished by the Government. Failures during flight testing (including equipment baselining) shall be analyzed for most probable cause. In those instances where the most probable cause is attributable to the radiosonde set the contractor will be required to take action as detailed in 4.5.3.2.

4.11 Interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.12, and in accordance with Gage List SC-GL-323214. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on spearable parts or assemblies which must mate, such as cases, covers and mounting.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assembly or major unit.

e. Size and form of special threads.

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4.12 Quality Conformance Inspection of Preparation for Delivery. - Preparation for delivery shall be inspected in accordance with MIL-R-11091.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing and Marking. - Preservation, packaging, packing and marking shall be in accordance with Specification MIL-R-11091.

6. NOTES

6.1 Intended Use. - The Radiosonde Set is attached to a free balloon and used to transmit data on air pressure, temperature, and humidity from the upper air to Rawin Set, AN/GMD-1 or AN/GMD-2, on the ground.

6.2 Ordering Data. - Procurement Documents should specify the following:

a. Title, number and date of this specification and any amendments thereto.

b. Type required.

c. Levels of preservation, packaging, and packing. (See Section 5)

d. If rough handling and functions tests are required.

e. Preproduction Inspection: Thirty sample units of each item cited in Section 1 (Including ancillary items as required) are generally required to reduce the overall time for lenghty environmental performance tests.

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical Literature required.

i. Statement of Treatment referenced in 3.4.14.2 to be submitted to contracting officer as soon as possible after award of contract.

6.3 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3)

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6.4 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, AN/AMT-12W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-55364 RADIOSONDE SET AN/AMT-12() (Units of)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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